

CSE437 Data Science-Coding With Real-World Data

Lecture-1

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[A part of this lecture is taken from the
lecture of Prof. John Canny of UC,
Berkeley]

“Data Science” an Emerging Field

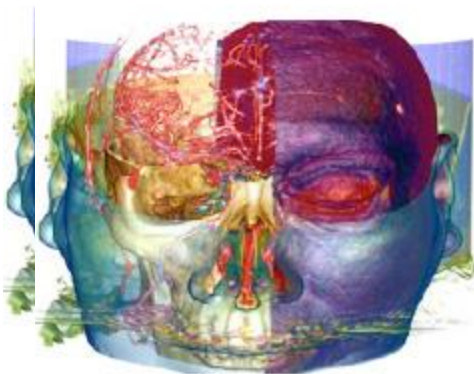


Data science is a multidisciplinary blend of **data inference, algorithm development, and technology** in order to solve analytically complex problems.

O'Reilly Radar report

Data Science

To gain insights into data through computation, statistics, and visualization



A Data Scientist Is...

“A data scientist is someone who knows more statistics than a computer scientist and more computer science than a statistician.”

- Josh Blumenstock

“Data Scientist = statistician + programmer + coach + storyteller + artist”

- Shlomo Aragmon

Phases of data science project

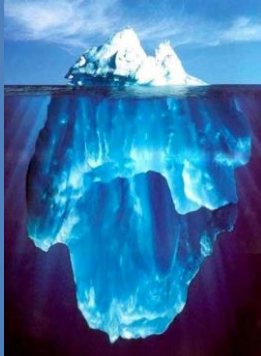
- 1) Business Problem: Asking question, objectives, parameters
- 2) Data acquisition: Databases, web servers, logs, API's, online repositories
- 3) Data preparation: (i) Data cleaning: inconsistent data and data-types, Misspelled attributes, Missing values, Duplicate values (ii) Transformation: mapping, quantization, binarization, normalization, scaling [professional tools: talend, informatica]

Phases of data science project

- 4) *Exploratory Data Analysis: Dimensionality reduction, Feature extraction, Feature selection
- 5) Data Modeling: ML (LR, LogR, KNN, DT, SVM, K-Means, Q-learning, NN, DNN, CNN, RNN, DBN, ANN, HMM, CRF, MEMM)-Train and Test to identify the best fit for business requirement.[Tools: Python, R, SAS]
- 6) Visualization and communication: Report generation and GUI [Professional tools: tableau, PowerBI, QlikView]
- 7) Deployment and maintenance: Initial deployment in a preproduction environment and then in the production environment. Get real-time analytics from report and dashboards. Control and Monitor project performance.

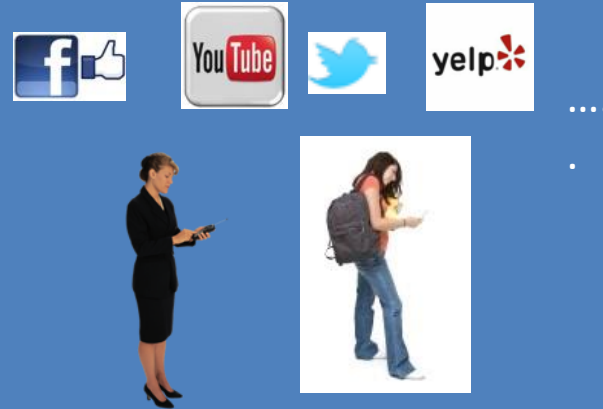
“Big Data” Sources

It's All Happening On-line



Every:
Click
Ad impression
Billing event
Fast Forward, pause,...
Server request
Transaction
Network message
Fault
...

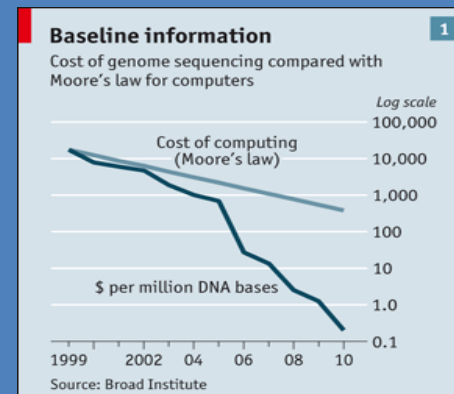
User Generated (Web & Mobile)



Internet of Things / M2M



Health/Scientific Computing

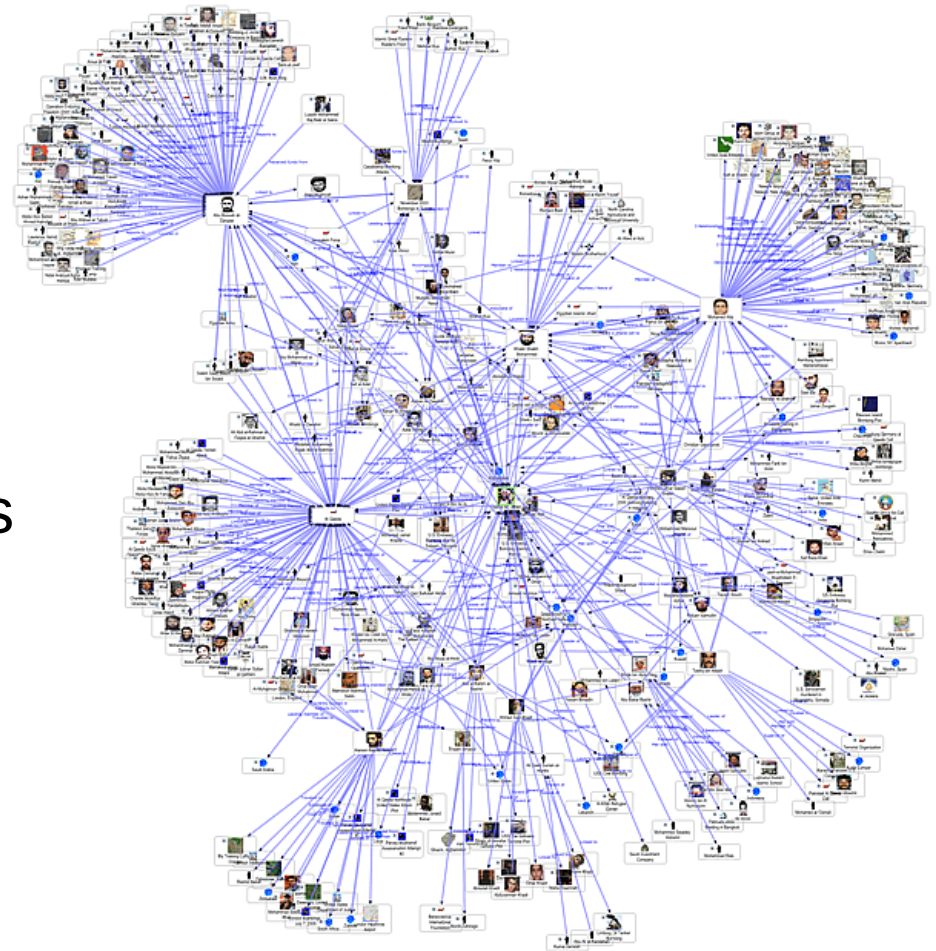


Graph Data

Lots of interesting data has a graph structure:

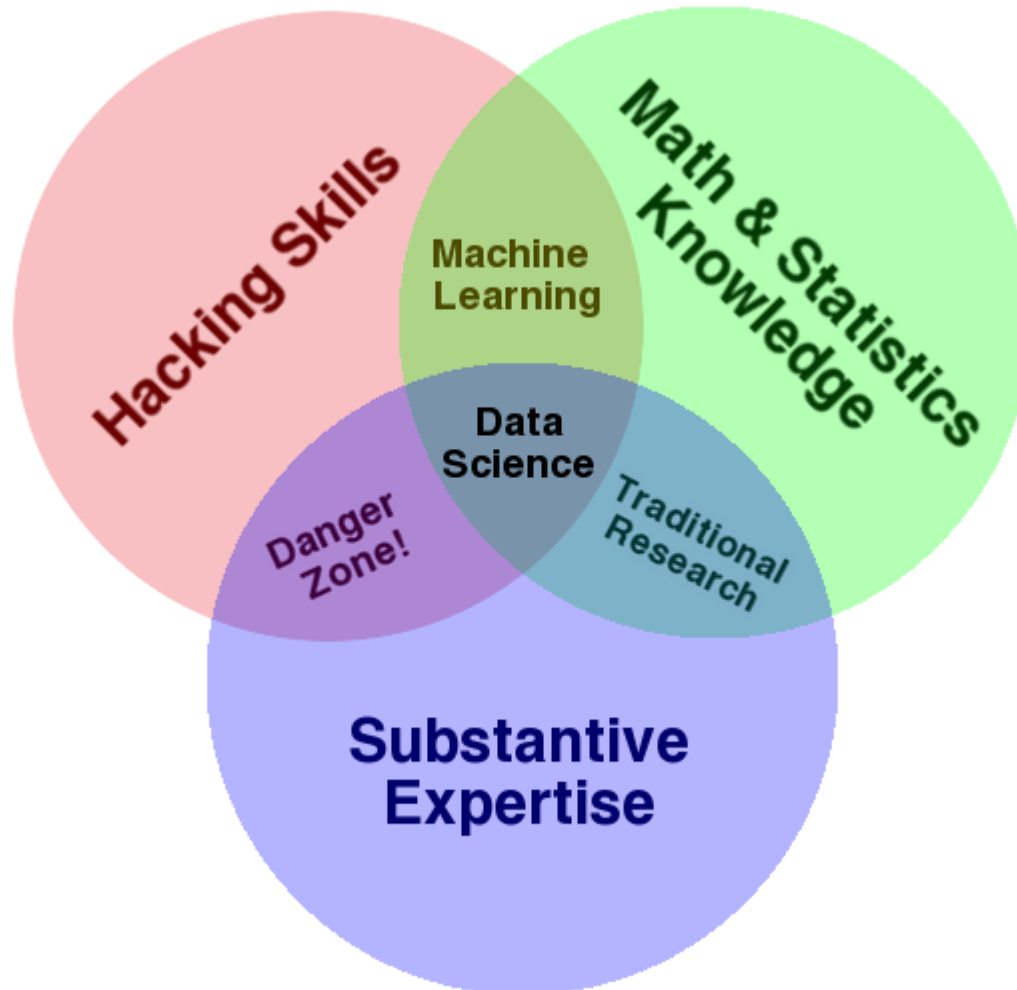
- Social networks
- Communication networks
- Computer Networks
- Road networks
- Citations
- Collaborations/Relationships
- ...

Some of these graphs can get quite large (e.g., Facebook* user graph)



DATA SCIENCE – WHAT IS IT?

Data Science – One Definition



Some recent DS/ML Competitions

Active Competitions				kaggle
		Flight Quest 2: Flight Optimization Final Phase of Flight Quest 2	33 days Coming soon \$220,000	
		Packing Santa's Sleigh He's making a list, checking it twice; to fill up his sleigh, he needs your advice	5.8 days 338 teams \$10,000	
		Flu Forecasting  Predict when, where and how strong the flu will be	41 days 37 teams	
		Galaxy Zoo - The Galaxy Challenge Classify the morphologies of distant galaxies in our Universe	2 months 160 teams \$16,000	
		Loan Default Prediction - Imperial College Lon... Constructing an optimal portfolio of loans	52 days 82 teams \$10,000	
		Dogs vs. Cats Create an algorithm to distinguish dogs from cats	11 days 166 teams Swag	

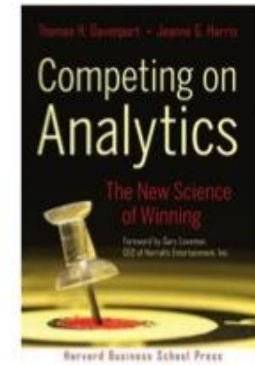
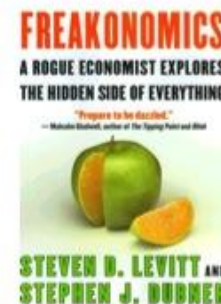
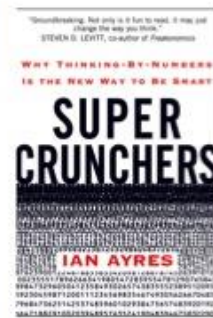
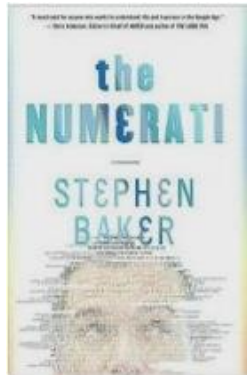
Contrast: Databases

	Databases	Data Science
Data Value	“Precious”	“Cheap”
Data Volume	Modest	Massive
Examples	Bank records, Personnel records, Census, Medical records	Online clicks, GPS logs, Tweets, Building sensor readings
Priorities	Consistency, Error recovery, Auditability	Speed, Availability, Query richness
Structured	Strongly (Schema)	Weakly or none (Text)
Properties	Transactions, ACID*	CAP* theorem (2/3), eventual consistency
Realizations	SQL	NoSQL: Riak, Memcached, Apache River, MongoDB, CouchDB, Hbase, Cassandra,...

ACID = Atomicity, Consistency, Isolation and Durability CAP = Consistency, Availability, Partition Tolerance

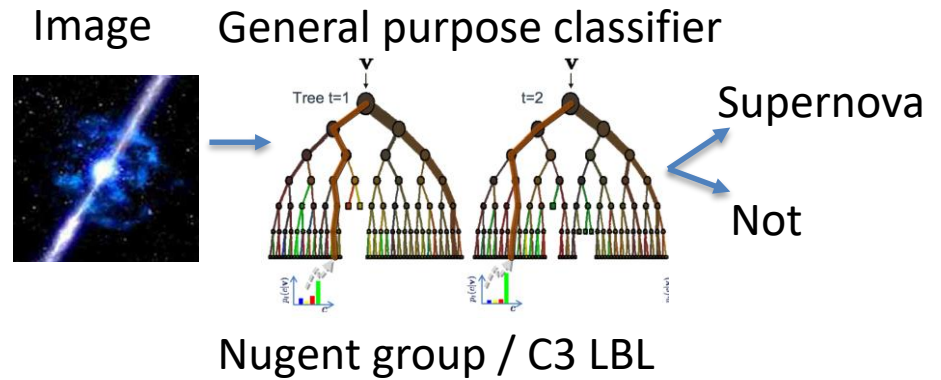
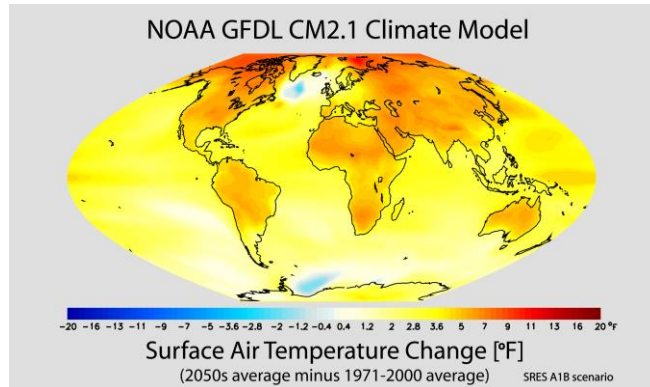
Contrast: Databases

Databases	Data Science
Querying the past	Querying the future



Business intelligence (BI) is the transformation of raw data into meaningful and useful information for [business analysis](#) purposes. BI can handle enormous amounts of unstructured data to help identify, develop and otherwise create new strategic business opportunities

Contrast: Scientific Computing



Scientific Modeling

Physics-based models

Problem-Structured

Mostly deterministic, precise

Run on Supercomputer or
High-end Computing Cluster

Data-Driven Approach

General inference engine replaces model

Structure not related to problem

Statistical models handle true randomness,
and **unmodeled complexity**.

Run on cheaper computer Clusters (EC2)

Contrast: Machine Learning

Machine Learning

Develop new (individual) models

Prove mathematical properties of models

Improve/validate on a few, relatively clean, small datasets

Publish a paper

Data Science

Explore many models, build and tune hybrids

Understand empirical properties of models

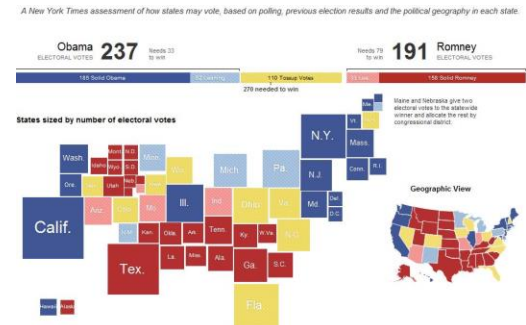
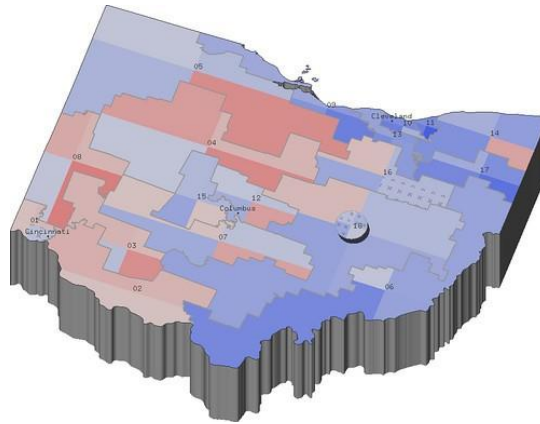
Develop/use tools that can handle massive datasets

Take action!

DOING DATA SCIENCE

Jeff Hammerbacher's Model

1. Identify problem
2. Instrument data sources
3. Collect data
4. Prepare data (integrate, transform, clean, filter, aggregate)
5. Build model
6. Evaluate model
7. Communicate results



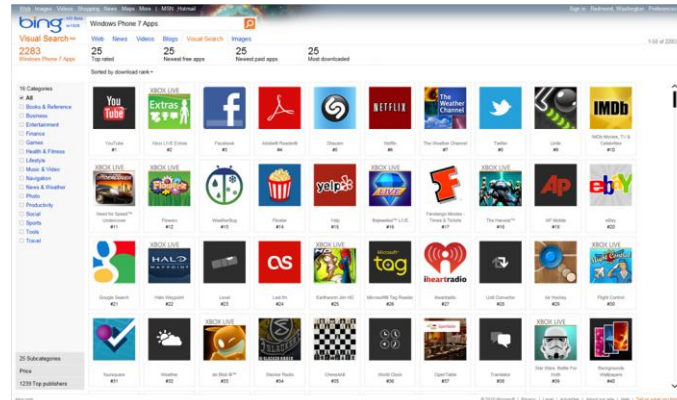
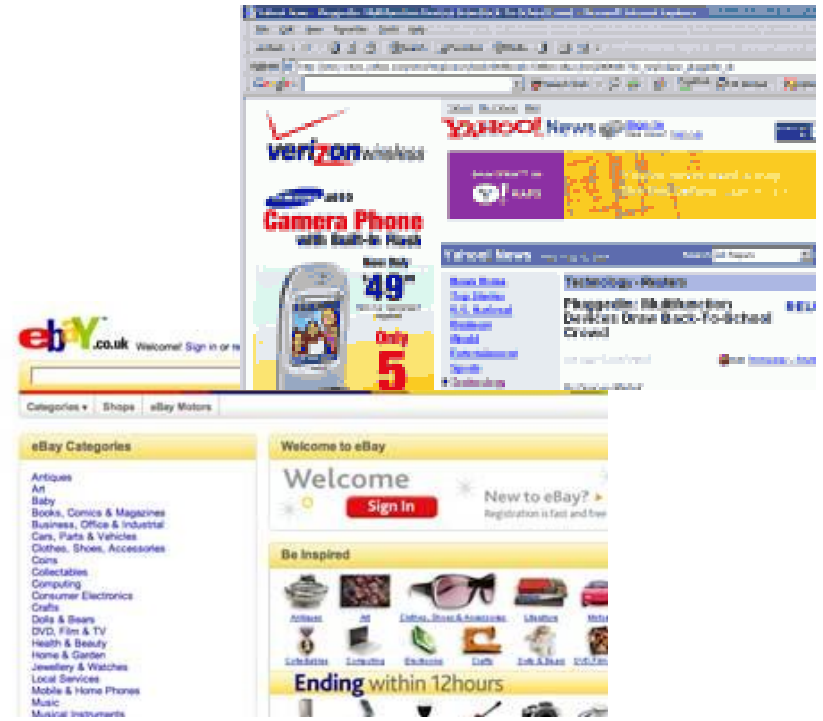
From the Trenches

Yahoo [KDD 2009, best app. paper]

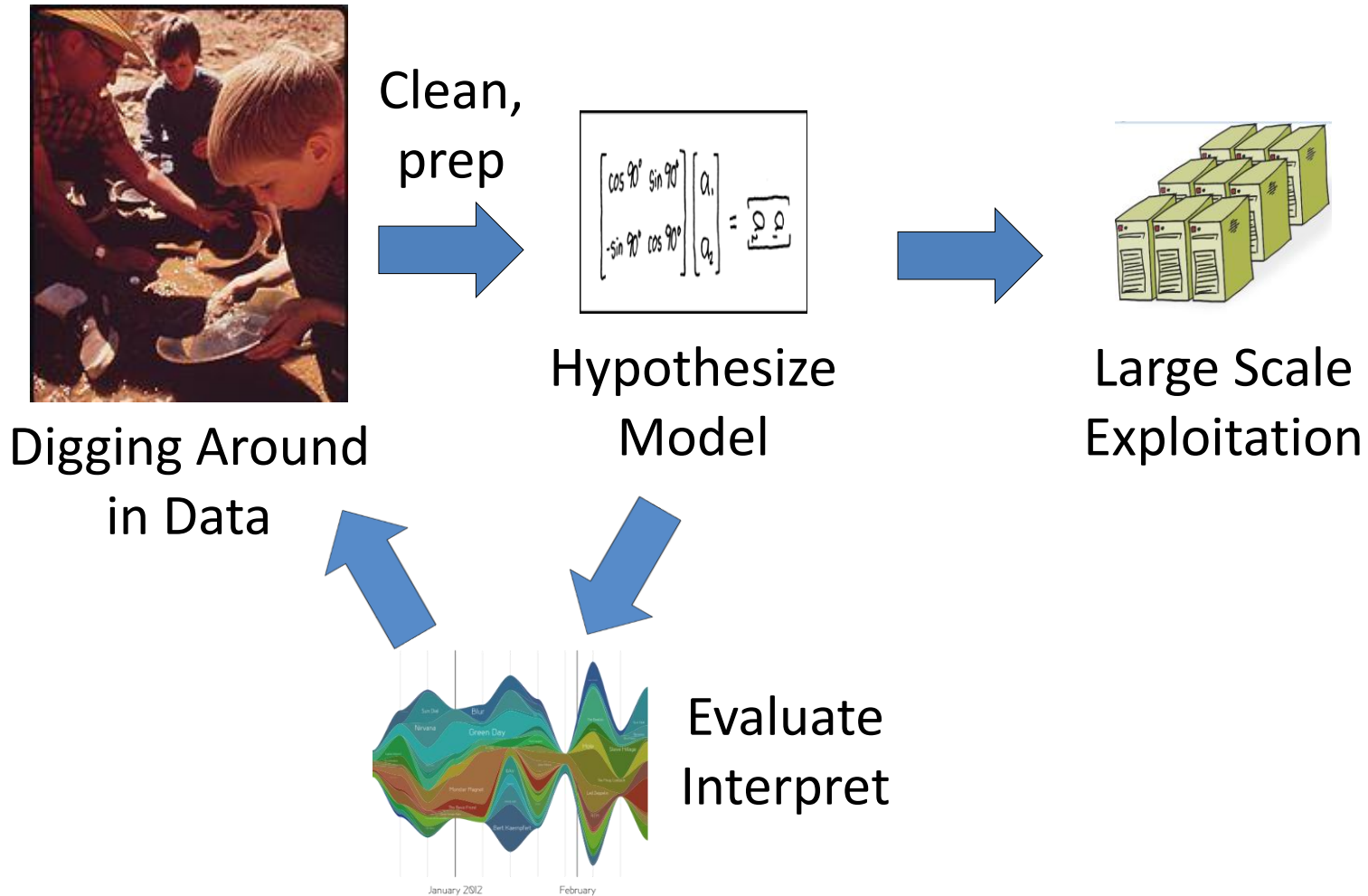
Ebay [SIGIR 2011, hon. mention]

Quantcast [2012]

Microsoft [CIKM 2014]



Data Scientist's Practice



What's Hard about Data Science

- Overcoming assumptions
- Making ad-hoc explanations of data patterns
- Overgeneralizing
- Communication
- Not checking enough (validate models, data pipeline integrity, etc.)
- Using statistical tests correctly
- Prototype → Production transitions
- Data pipeline complexity